

MedTrack_DV (Hospital Operations & Patient Analytics Dashboard)

Project Statement: This project aims to develop a comprehensive hospital operations and patient analytics dashboard suite to analyse hospital performance, patient admissions, department efficiency, and healthcare resource utilisation. Using publicly available Hospital Datasets and Patient Admission Datasets, the project will transform healthcare operational data into actionable insights through interactive Tableau dashboards. The dashboard suite will help hospital administrators, healthcare managers, medical staff, and policymakers monitor patient flow, evaluate department performance, optimise hospital resources, and improve operational efficiency through data-driven decision-making.

Outcomes:

- Collection and integration of hospital datasets.
- Data cleaning and transformation for healthcare operations analytics.
- Development of hospital performance KPIs.
- Creation of four interactive Tableau dashboards.
- Patient flow and admission analysis.
- Department efficiency monitoring.

Milestone 1: Data Collection and Preparation Module 1: Hospital Data Collection

Dataset Resources:

Dataset Name: Hospital HMIS Dataset for Healthcare Analytics

Dataset Description: This dataset represents a synthetic but realistic Hospital Management Information System (HMIS) designed for healthcare analytics, dashboards and operational insights. It contains multiple interconnected tables covering patient information, admissions, departments, wards, beds, billing, insurance and staff management.

The dataset is designed to represent the functioning of a hospital management information system. The **admission table acts as the central transactional table**, which can be connected with patient, department, ward, disease and billing-related tables using common keys.

Dataset Link: <https://www.kaggle.com/datasets/shalakagangurde/hospital-hmis-dataset-for-healthcare-analytics>

Source: Kaggle

Tables Used in This Project:

For this project, the tables that were useful for hospital operations analysis were selected and integrated into a consolidated dataset.

- **Patient table** – Used for patient details and demographics.
- **Admission table** – Used for admission, discharge, patient, department, ward and disease details.
- **Department table** – Used for department names and analysis.
- **Ward table** – Used for ward details, ward types and bed capacity.
- **Bed table** – Used for bed information and resource utilisation.

- **Disease table** – Used for disease names and categories.
- **Billing table** – Used for billing, insurance and patient payable amounts.
- **Employee table** – Used for employee details and staff count.
- **Staff Assignment table** – Used for department/ward-wise staff allocation and shifts.

Data Integration:

The selected tables were integrated using their common key fields. The Admissions table was used as the primary transaction table, while the Patients, Departments, Wards, Beds, Disease, Billing, Employee, and Staff Assignment tables were connected through their respective identifiers.

The main relationships used during integration were:

- patient_id connects Patients and Admissions
- department_id connects Admissions, Departments, Wards, and Employee
- ward_id connects Admissions, Wards, Beds, and Staff Assignment
- bed_id connects Admissions and Beds
- disease_id connects Admissions and Disease
- admission_id connects Admissions and Billing
- employee_id connects Employee and Staff Assignment

The Integrated Dataset has 45000 rows and 34 columns. This dataset has been used in further analysis, such as cleaning and transformation.

Deliverables

- **DataCollection.py**
- **Hospital_raw_data.xlsx**

Module 2: Data Cleaning & Transformation

Data Cleaning & Transformation:

The original dataset contained **45,000 rows and 34 columns**, with no duplicate records. However, to practice and demonstrate data-cleaning techniques, I manually introduced data inconsistencies, including **2% duplicate rows, 1% missing values, 5% inconsistent department names, 2% inconsistent healthcare indicators, and 2% blood group values converted to lowercase**. After introducing these issues, appropriate data-cleaning techniques were applied, such as duplicate removal, missing-value handling, standardisation of department names, correction of healthcare indicators, and formatting blood group values consistently. Required transformations and calculated fields were then created to prepare the final dataset for KPI calculation and dashboard development.

Data Cleaning Techniques

Data Cleaning is the process of identifying and correcting inaccurate, incomplete, duplicate, or inconsistent data. It was used to improve data quality, maintain consistency, and ensure accurate KPI calculations and dashboard results.

1. **Duplicate Removal** – Removed duplicate records to avoid repeated data and incorrect KPI calculations.
2. **Missing Value Handling** – Identified and handled missing values to prevent errors during analysis.
3. **Department Name Standardisation** – Standardised inconsistent department names for accurate department-wise analysis.
4. **Healthcare Indicator Correction** – Corrected inconsistent values in healthcare indicators such as admission type and admission status.

5. **Blood Group Standardisation** – Converted lowercase blood group values into a consistent format for accurate filtering and grouping.
6. **Data Type Correction** – Converted dates, numerical values and categorical fields into appropriate data types.

Data Transformation Techniques

Data Transformation is the process of converting and organising cleaned data into a suitable format for analysis. It was used to create calculated fields, categorise data, integrate tables, and prepare meaningful information for dashboard visualisation.

1. **Date Transformation** – Converted admission and discharge dates into proper date formats for monthly and yearly analysis.
2. **Age Group Creation** – Created age groups to analyse patient demographics.
3. **Length of Stay Calculation** – Calculated the number of days between admission and discharge dates.
4. **Data Aggregation** – Summarised data by department, ward, disease category and month.
5. **Financial Transformation** – Prepared total amount, insurance-covered amount and patient-payable amount for financial analysis.

The Cleaned Dataset is used in further analysis for KPI's Creation and Dashboard Development.

Deliverables

- hospital_cleaned.csv
- hospital_cleaning.ipynb

Module 3: Hospital KPI Engineering

KPI Calculations:

A **Key Performance Indicator (KPI)** is a measurable value used to evaluate performance, efficiency and progress of an organisation or process. KPIs help decision-makers understand the current situation, identify problems and make better decisions based on data.

How KPIs Are Useful in Our Project

Our project is a Hospital Operations & Patient Analytics Dashboard. The purpose of our project is to analyse hospital and patient data and provide useful insights through Python, SQL and Tableau. KPIs are useful because they convert large amounts of hospital data into simple numbers that management can understand and monitor. In our hospital analytics project, KPIs monitor key healthcare operations and measure hospital performance. They provide a quick summary of key information through dashboard cards and help hospital management identify areas that need improvement.

KPI's Used in This Project:

1. **Total Admissions** – Measures the total number of patient admissions and helps understand hospital workload.

Formula: Total Admissions = COUNT(Admission_ID)

2. **Total Patients** – Shows the number of unique patients treated in the hospital.

Formula: Total Patients = COUNT(DISTINCT Patient_ID)

3. **Total Discharges** – Measures the number of patients discharged and helps analyse patient flow.

Formula: Total Discharges = COUNT(DISTINCT Admission_ID), where Discharge_Date is not null

4. **Occupancy Rate** – Shows the percentage of hospital beds occupied by patients and helps monitor bed availability.
Formula: Occupancy Rate (%) = (Number of Occupied Beds / Total Beds) × 100
6. **Readmission Rate** – Measures how frequently patients are admitted again and helps identify possible quality-of-care issues.
Formula: Readmission Rate (%) = (Number of Patients with More Than One Admission / Total Patients) × 100
7. **Average Length of Stay** – Shows the average number of days patients stay in the hospital and helps evaluate treatment efficiency and bed usage.
Formula: Average Length of Stay = SUM(Discharge_Date – Admission_Date) / Number of Discharged Patients
8. **Bed Utilisation Rate** – Measures how effectively hospital beds are being used.
Formula: Bed Utilisation Rate (%) = (Number of Patients / Total Bed Capacity) × 100
9. **Department Efficiency** – Helps compare department performance based on admissions, discharges, resources and operational activities.
Formula: Department Efficiency (%) = (Department Output / Department Resources) × 100
10. **Total Revenue** – Shows the total billing amount generated by the hospital.
Formula: Total Revenue = SUM(Patient_Payable_Amount + Insurance_Covered_Amount)
11. **Total Staff** – Measures the number of employees available for hospital operations.
Formula: Total Staff = COUNTD(Employee_ID)
12. **Total Wards** – Shows the total number of wards available in the hospital.
Formula: Total Wards = COUNTD(Ward_ID)
13. **Total Beds** – Measures the total bed capacity of the hospital.
Formula: Total Beds = SUM(Total_Beds)
14. **Average Beds per Ward** – Shows the average number of beds available in each ward and helps evaluate ward capacity.
Formula: Average Beds per Ward = Total Beds / Total Wards

Overall, these KPIs help hospital management monitor performance, improve resource utilisation, identify operational issues and make data-driven decisions.

Deliverables

- hospital_final_dataset.xlsx
- generate_hospital_kpis.py

Module 4: Dashboard Planning & Prototyping

Design layouts for: Hospital Overview

Patient Flow
 Department Analytics
 Resource Utilization
 Filters Navigation
 Dashboard actions
 Department comparisons

Deliverables

- dashboard_storyboard.pdf, medtrack_prototype.twbx

Module 5: Build Hospital Overview & Patient Flow Dashboards

Dashboard Guide:

Dashboards: Hospital Overview

Patient Flow

1. Hospital Overview Dashboard

Purpose: The Hospital Overview Dashboard provides a high-level view of hospital performance and operations. It helps management understand admission trends, bed utilisation, occupancy, readmission levels, and department efficiency.

Filters Used:

- **Admission Month:** Used to view hospital performance for a selected month or time period.
- **Admission Type:** Used to analyse different types of patient admissions.
- **Department Name:** Used to view the performance of a specific department.
- **Ward Type:** Used to compare admissions and resource usage across different ward types.
- **Age Group:** Used to analyse admissions and discharges based on patient age categories.

KPIs Used:

- **Total Admissions:** Shows the total number of patient admissions and helps measure hospital workload.
- **Occupancy Rate:** Shows the percentage of beds occupied and helps monitor bed availability.
- **Readmission Rate:** Measures the percentage of patients admitted again and helps identify possible quality-of-care issues.
- **Bed Utilisation Rate:** Shows how effectively hospital beds are being used.
- **Department Efficiency:** Helps compare the operational performance of different departments.

Charts Used and Their Purpose:

- **Monthly Admission Trend:** Used to identify admission patterns and changes over time.
- **Admissions by Admission Type:** Used to compare admission categories and understand patient inflow.
- **Admissions by Ward Type:** Used to analyse how patients are distributed across different ward types.
- **Admissions by Age Group:** Used to understand which age groups contribute most to hospital admissions.
- **Discharges by Age Group:** Used to compare the number of discharged patients across different age groups.
- **Total Revenue by Departments:** Used to compare revenue generated by each department and identify high-revenue departments.
- **Average LOS by Department:** Used to compare the average length of patient stay across departments and identify departments with longer or shorter stays.

Key Insights:

The dashboard helps identify admission trends, understand patient demographics, monitor occupancy and bed utilisation, compare department revenue, and analyse average patient stay. These insights support better resource planning, patient-flow management, and operational decision-making.

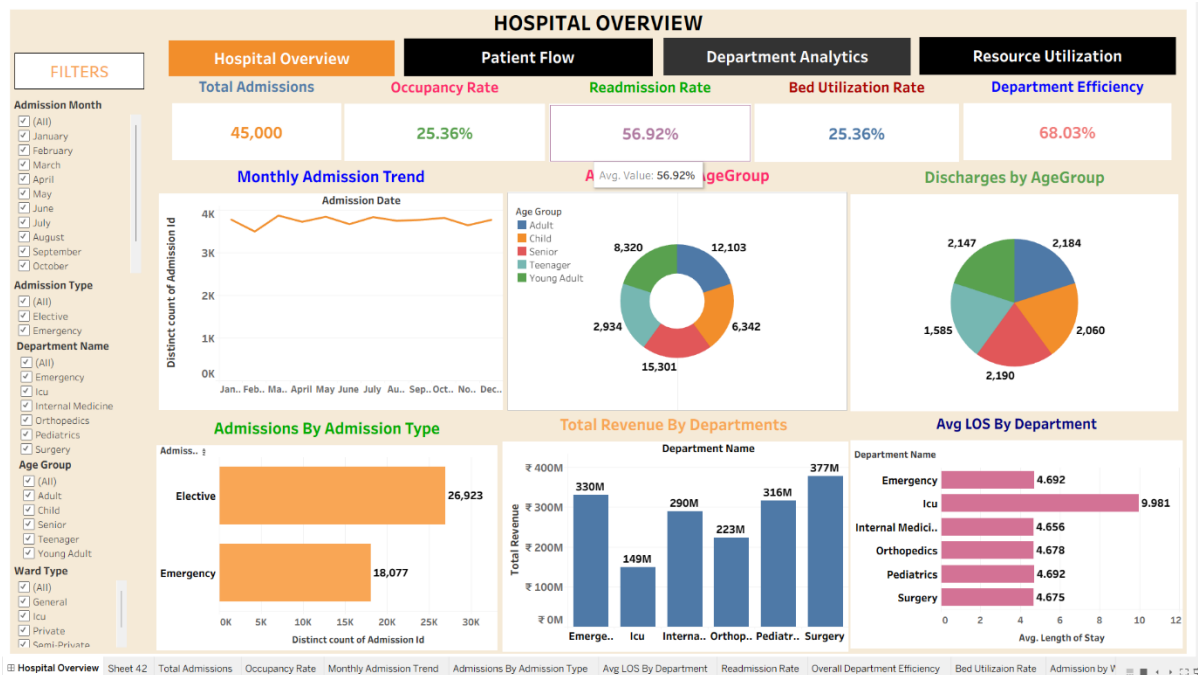


Fig-1 Hospital Overview

2. Patient Flow Dashboard

Purpose:

The Patient Flow Dashboard analyses the movement of patients through the hospital, from admission to discharge. It helps management understand patient volume, discharge trends, department-wise patient distribution, disease patterns, ward utilisation, and readmission levels.

Filters Used:

- **Admission Type:** Used to analyse elective and emergency admissions.
- **Department Name:** Used to view patient flow for a particular department.
- **Ward Type:** Used to compare patient distribution across different ward types.
- **Age Group:** Used to analyse patient flow across different age categories.
- **Blood Group:** Used to filter patients based on their blood group.

KPIs Used:

- **Total Patients:** Shows the number of unique patients treated in the hospital.
- **Total Discharges:** Shows the number of discharged patients.
- **Average Length of Stay:** Shows the average number of days patients stay in the hospital.
- **Readmission Rate:** Measures the percentage of patients admitted again.
- **Emergency Admission Rate:** Shows the percentage of admissions that were emergency cases.

Charts Used and Their Purpose:

- **Discharges by Monthly Trend:** Used to identify monthly discharge patterns and changes over time.
- **Patients by Department:** Used to compare the number of patients across departments.
- **Patients by Disease Category:** Used to understand the distribution of patients based on disease categories.
- **Patients by Ward Type:** Used to compare patient distribution across General, ICU, Private, and Semi-Private wards.
- **Admissions by Disease Name:** Used to identify the diseases contributing to patient admissions.
- **Discharges by Ward Type:** Used to compare discharge volumes across different ward types.

Key Insights:

The dashboard helps identify patient admission and discharge patterns, departments with higher patient volumes, common disease categories, and ward-wise patient distribution. It also supports monitoring of average patient stay, emergency admissions, and readmission levels for better patient-flow management.

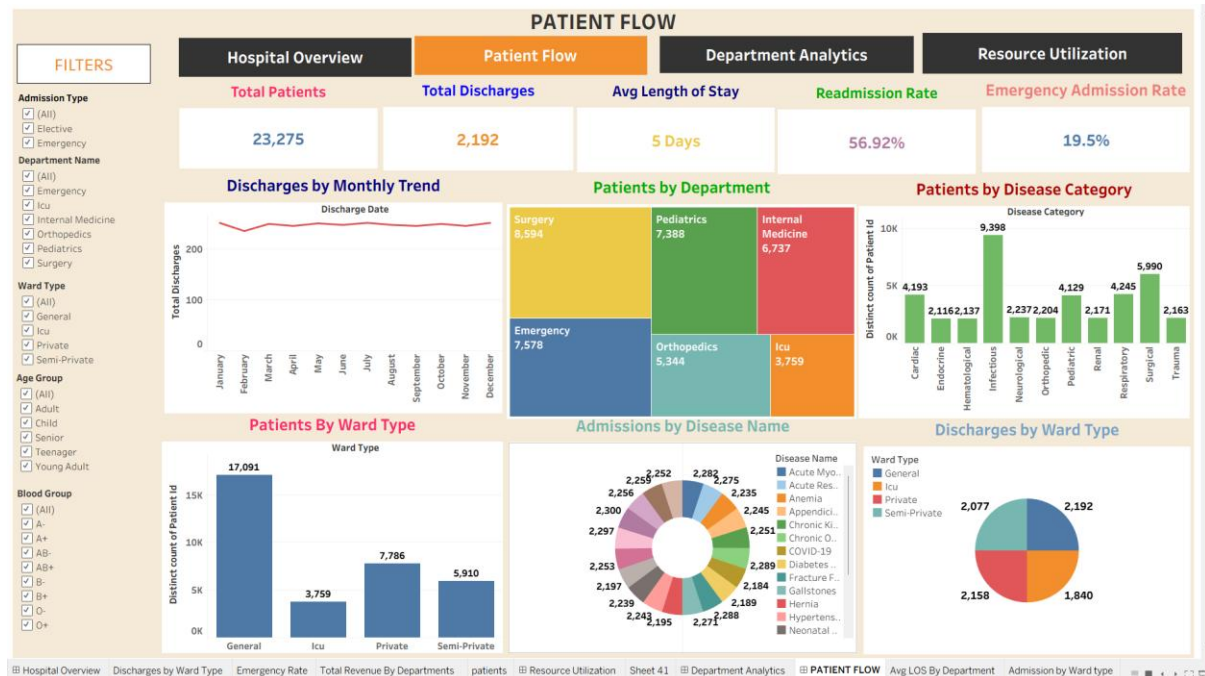


Fig-2 Patient Flow

Opening the Workbook

1. Open dashboard/Hospital Operations and Patient Analysis.twbx in Tableau Desktop.
2. If Tableau reports a missing file, use Edit Connection and select data/hospital_cleaned .csv.
3. Refresh the data source and confirm that the fields load without errors.
4. Start with Hospital Overview for overall context, then move to the dashboard that matches the question.

Recommended Workflow

1. Select a hospital type, city, or hospital.
2. Narrow the period using admission year or month where available.
3. Select a department to investigate capacity, staffing, or patient flow.
4. Read the KPI cards before interpreting charts.
5. Select marks in charts to use the workbook's highlight and filter actions.
6. Clear selections before starting a new comparison.

Deliverables

medtrack_dashboard_v1.twbx

Module 6: Build Department Analytics & Resource Utilisation Dashboards

Dashboard Guide:

Dashboards: Department Analytics

Resource Utilization

3. Department Analytics Dashboard

Purpose:

The Department Analytics Dashboard compares the performance of different hospital departments. It helps management evaluate department-wise patient volume, efficiency, revenue, readmission rates, and resource requirements.

Filters Used:

- **Discharge Date:** Used to analyse department performance for a selected discharge period.
- **Disease Category:** Used to filter department analysis based on disease categories.
- **Ward Type:** Used to compare department performance across different ward types.
- **Disease Name:** Used to analyse departments based on specific diseases.

KPIs Used:

- **Total Departments:** Shows the number of departments included in the analysis.
- **Total Patient Payable:** Shows the total amount payable by patients.
- **Total Insurance Covered:** Shows the total amount covered by insurance.
- **Total Beds:** Shows the total bed capacity associated with the selected departments.
- **Total Revenue:** Shows the total revenue generated by the hospital.

Charts Used and Their Purpose:

- **Department Patients by Disease Category:** Used to compare patient counts across departments and disease categories.
- **Efficiency Score by Department:** Used to compare the operational efficiency of different departments.
- **Monthly Admissions by Department:** Used to analyse admission trends for each department over time.
- **Discharges by Department:** Used to compare the number of discharged patients across departments.
- **Readmission Rate by Department:** Used to identify departments with higher or lower readmission rates.
- **Patients Payable by Department:** Used to compare the patient-payable amount generated by each department.

Key Insights:

The dashboard helps identify departments with higher patient volumes, better efficiency scores, greater revenue contribution, and higher readmission rates. It also supports department-level resource planning, financial analysis, and performance improvement.

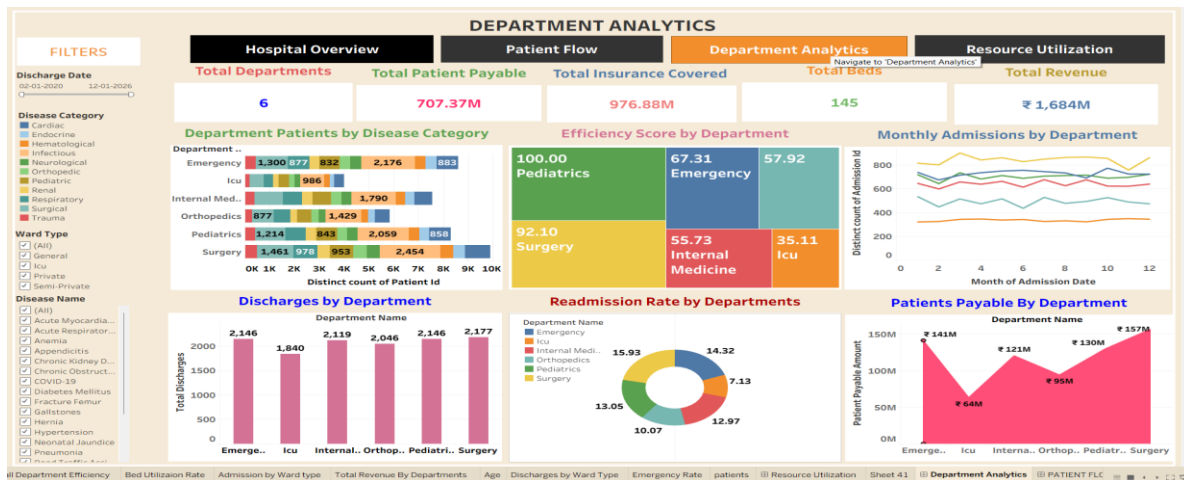


Fig-3 Department Analytics

4. Resource Utilisation Dashboard

Purpose:

The Resource Utilisation Dashboard evaluates how effectively hospital resources such as beds, wards, and staff are being used. It helps management monitor resource availability, bed utilisation, staff allocation, and department-wise resource requirements.

Filters Used:

- **Admission Type:** Used to analyse resource utilisation based on admission categories.
- **Age Group:** Used to understand resource usage across different patient age groups.
- **Ward Type:** Used to compare resource utilisation across different ward types.
- **Department Name:** Used to analyse resource usage for a specific department.
- **Disease Category:** Used to examine resource utilisation based on disease categories.

KPIs Used:

- **Total Beds:** Shows the total number of beds available in the hospital.
- **Total Staff:** Shows the total number of hospital employees.
- **Total Wards:** Shows the total number of wards.
- **Average Beds per Ward:** Shows the average number of beds available in each ward.
- **Bed Utilisation Rate:** Measures how effectively hospital beds are being used.

Charts Used and Their Purpose:

- **Discharges by Disease Category:** Used to analyse discharge volumes across different disease categories.
- **Total Beds by Department:** Used to compare bed capacity across departments.
- **Beds by Ward Type:** Used to understand how beds are distributed among different ward types.
- **Average Staff Allocation by Department:** Used to compare staff allocation across departments.
- **Bed Utilisation by Department:** Used to monitor bed utilisation trends for each department over time.
- **Department-wise Payable by Age Group:** Used to compare patient-payable amounts across departments and age groups.

Key Insights:

The dashboard helps identify departments and wards with higher bed capacity, compare staff allocation, monitor bed utilisation trends, and understand resource requirements based on patient demographics and disease categories. These insights support better resource planning, capacity management, and operational efficiency.

Deliverables

MedTrack_DV.twbx

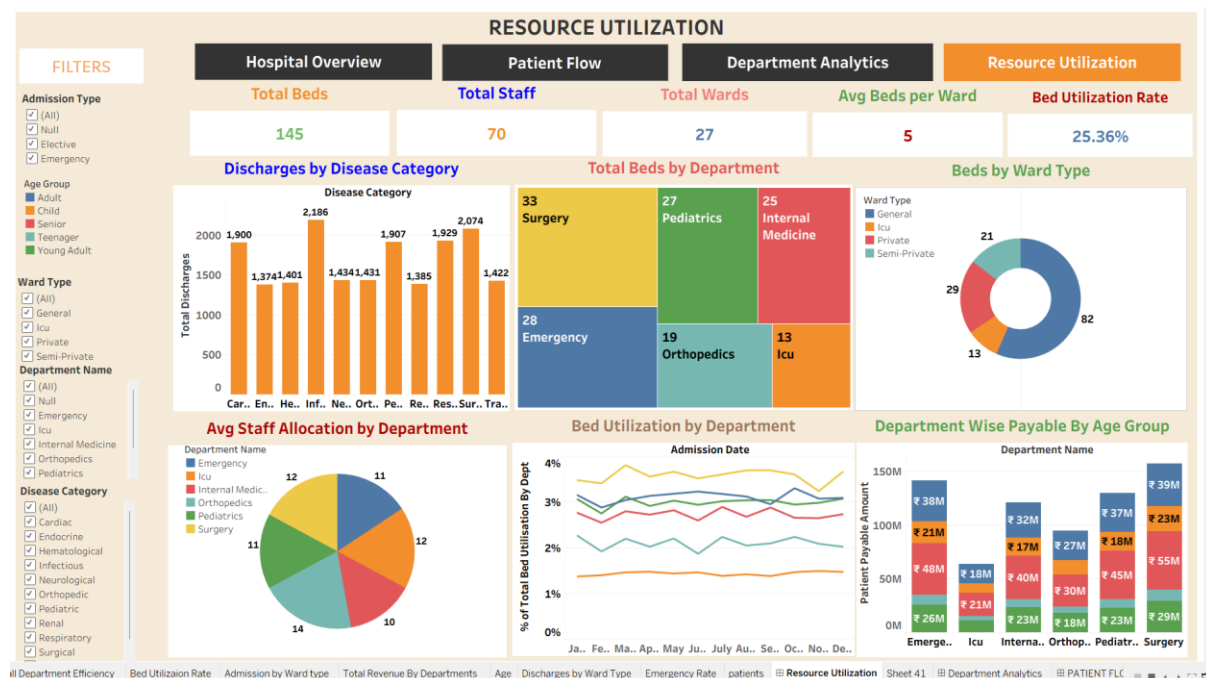


Fig-4 Resource Utilization

Module 7: Testing and Validation

1. Objective

The objective of this testing report is to validate the accuracy, reliability, usability and consistency of the final MedTrack Tableau hospital analytics dashboards. Testing covered KPI calculations, healthcare metrics, patient-flow analysis, department-level analysis, resource-utilization analysis, filters, dashboard navigation, chart interactions and data-quality checks.

2. Dataset Used for Testing

The dashboards were developed using an integrated and cleaned hospital healthcare dataset. The final integrated dataset contains 45,000 records and 34 columns. The source data was combined from relevant hospital entities such as patients, admissions, departments, wards, beds, billing, staff assignments, employees and disease information.

Testing Methodology

- Testing was performed by comparing the dashboard outputs with the cleaned integrated dataset and independently calculated aggregations. KPI definitions were checked against the project requirements. Dashboard filters, cross-filtering, navigation buttons, chart labels, tooltips and visual consistency were tested manually.
- Verify that the dashboard uses the correct data source and fields.
- Compare displayed KPI values with independent calculations from the final dataset.
- Check date, department, admission-type, age-group, blood-group and ward-related filters.
- Verify that charts respond correctly when filters are applied.
- Check that navigation buttons open the intended dashboard.
- Check chart titles, legends, labels, tooltips and readability.
- Review duplicate records and missing-value issues that may affect calculations.

KPI and Data Validation Summary

Validation Area	Expected Result	Observed Result	Status
Total admissions	Based on the integrated admissions dataset	Dashboard aggregation checked	PASS
Total patients	Distinct patient identification	Dashboard aggregation checked	PASS
Total discharges	Records with valid discharge information	Dashboard aggregation checked	PASS
Average LOS	Calculated from admission and discharge dates	Dashboard value checked	PASS
Readmission rate	Patients with more than one admission	Dashboard calculation checked	PASS
Bed utilization rate	Based on the selected project definition	Dashboard calculation checked	PASS
Department efficiency	Department-level efficiency calculation	Dashboard calculation checked	PASS
Total beds and wards	Distinct/aggregated resource counts	Dashboard aggregation checked	PASS

Data Quality Testing

- The final integrated dataset was checked for duplicate records affecting dashboard results.
- Admission and discharge dates were reviewed for length-of-stay calculations.
- Missing discharge dates were considered when calculating discharge-related measures.
- Department, ward, bed and patient relationships were checked for consistency.
- The dashboard was checked for incorrect aggregation caused by duplicated relationship records.
- Chart totals, KPI totals and filter results were reviewed for consistency.

Conclusion

The final MedTrack Tableau hospital analytics dashboards successfully passed the testing and validation process. The four dashboards—Hospital Overview, Patient Flow, Department Analytics and Resource Utilization—were checked for KPI accuracy, chart correctness, filter functionality, cross-filtering, navigation and presentation quality. The dashboards provide a consistent and interactive view of hospital operations, patient flow, department performance and resource utilization.

Module 8: Documentation and Project Delivery

Prepared project documentation including:

Dataset sources

KPI definitions

Dashboard guide

Healthcare operations methodology

Organised project structure: /scripts /data /dashboard /docs

Tech Stack Used:

Area	Tools / Libraries
Data Collection	Python, Hospital Dataset, Patient Admission Dataset
Data Processing	Pandas, NumPy
Data Cleaning	Python
Visualization	Tableau Desktop / Tableau Public
Dashboard Integration	Tableau Filters, Parameters, Actions
Documentation	Markdown, GitHub